

Sunny and shadow fluctuations - Experiment description

OS4P

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1 Introduction

In this experiment, the aim was to determine the effect of the sun on atmospheric conditions. The experiment was performed in the Botanical Gardens at Utrecht University. The parameters measured for this aim were pressure(Pa), temperature(C) and humidity(%) using an experimental set-up consisting of an Arduino Uno and a P/T/RH sensor. The experimental values are then compared to the weather report for that day.

2 Methods

2.1 Experimental set-up

Components used in the set up: Breadboard, Arduino Uno, Cables, P/T/RH sensor(BME280), SD card, SD card reader and External battery (powerbank).

Pictures/ Scheme arduino The Arduino is plugged to a computer from which it can be controlled using a code. The code is made with Arduino IDE.

2.2 Collecting data

The data was acquired using the aforementioned experimental set-up and the scripts: "Erase SD card", "Recording data on P,T,RH sensor" and "Extract data from csv file", downloadable from Experiment repository.

The scripts should be run in the following order:

1. WITHOUT THE P/T/RH sensor(BME280) connected to breadboard, check that the SD card is working properly by running "Extract data from csv file". If all is fine "Initializing SD card... SUCCESS!" will be printed, if not "Initializing SD card... FAILED!" will be printed.
2. If SD card is working properly, erase existing data from SD card by running the "Erase SD card" script.

3.

3 Results

plots

4 Discussion

Compare plot with weather report.

5 Conclusion